Problem Description | Solution | Video | Comments 0 | Playground

Matrix-Vector Dot Product

Easy | Linear Algebra

Write a Python function that computes the dot product of a matrix and a vector. The function should return a list representing the resulting vector if the operation is valid, or -1 if the matrix and vector dimensions are incompatible. A matrix (a list of lists) can be dotted with a vector (a list) only if the number of columns in the matrix equals the length of the vector. For example, an n x m matrix requires a vector of length m.

Example:

Input:

```
a = [[1, 2], [2, 4]], b = [1, 2]
```

Output:

```
[5, 10]
```

Reasoning:

Row 1: (1 * 1) + (2 * 2) = 1 + 4 = 5; Row 2: (2 * 1) + (4 * 2) = 2 + 8 = 10

Learn About topic

Contributors:

Moe Chabot

Contribute

Request Edit

Notebook Mode | NumPy

```
# This needs to be imported at the top of the file, inside the Deep-ML code writing section.
import numpy as np


def matrix_dot_vector(a: list[list[int | float]], b: list[int | float]) -> list[int | float]:
    # Return a list where each element is the dot product of a row of 'a' with 'b'.
    # If the number of columns in 'a' does not match the length of 'b', return -1.

    # Initially converting the python list of lists into a numpy array. But we'll not change the data type at this point.
    a = np.array(a)
    b = np.array(b)

    # Checking if the number of columns in 'a' matches the length of 'b'.
    if np.shape(a)[1] != len(b):
        # If not, returning -1.
        return -1
    else:
        # If yes, calculating the dot product of each row of 'a' with 'b'.
        ab_dot_product = np.dot(a, b)

```

Run Code | Submit Code | Reset | Save Code | Submissions 2

Case 1 +

Test Code

```
print(matrix_dot_vector([[1, 2, 3], [2, 4, 5], [6, 8, 9]], [1, 2, 3]))
```

Expected Output

```
[14, 25, 49]
```

Test Results 8/8

Test 1 | Test 2 | Test 3 | Test 4 | Test 5 | Test 6 | Test 7 | Test 8

Test Case

```
print(matrix_dot_vector([[1, 2, 3], [2, 4, 5], [6, 8, 9]], [1, 2, 3]))
```

Expected	Your Output	Passed
[14, 25, 49]	[14, 25, 49]	

Ask Tutor